

# DREF MANUAL

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[in package DREF]

## 1 Links and Systems

The official repository is <https://github.com/melisgl/dref>, and this document is available in various formats on <https://fixnum.com> for the latest version.

- **[system]** `"dref"`
  - *Version:* 0.5.0
  - *Description:* Reify definitions, provide portable access to arglists, docstrings and source locations in an extensible framework.
  - *Long Description:* `defun` defines a first-class object: a `function`. `defvar` does not. The DRef library provides a way to refer to all definitions and smooths over the differences between implementations. This system has minimal dependencies. It autoloads the `dref/full` `asdf:system`.
  - *Licence:* MIT, see COPYING.
  - *Author:* Gábor Melis
  - *Mailto:* [mega@retes.hu](mailto:mega@retes.hu)
  - *Homepage:* <http://github.com/melisgl/dref/>
  - *Bug tracker:* <https://github.com/melisgl/dref/issues>
  - *Source control:* `GIT`
  - *Depends on:* `autoload`, `mgl-pax-bootstrap`, `named-readtables`, `pythonic-string-reader`
  - *Auto depends on:* `dref/full`
  - *Defsystem depends on:* `autoload`
- **[system]** `"dref/full"`
  - *Description:* DRef with everything loaded. There should be no need to explicitly load this system (or depend on it) because it is autoloaded as necessary by all publicly accessible functionality in `dref`.

However, to get the dependencies, install this system.

For a discussion of conditional dependencies, see [Backends](#).

- *Depends on:* alexandria, closer-mop, [dref](#), mgl-pax, sb-introspect(?), swank(?)
- *Defsystem depends on:* autoload

## 2 Introduction

*What if definitions were first-class objects?*

Some **defining forms** do not create first-class **objects**. For example, **defun** creates **function** objects, but **defvar** does not create variable objects, as no such thing exists in Common Lisp. The main purpose of this library is to fill this gap with the introduction of **dref** objects:

```
(defvar *my-var* nil
  "This is my var.")
(dref '*my-var* 'variable)
==> #<DREF *MY-VAR* VARIABLE>
```

drefs just package up a **name** (*\*my-var\**) and a **locative** (*variable*) then check that the definition actually exists:

```
(dref 'junk 'variable)
.. debugger invoked on LOCATE-ERROR:
.. Could not locate JUNK VARIABLE.
```

The **Basic Operations** on definitions in DRef are **arglist**, **docstring** and **source-location**.

```
(docstring (dref '*my-var* 'variable))
=> "This is my var."
=> NIL
```

For definitions associated with objects, the definition can be **located** from the object:

```
(locate #'print)
==> #<DREF PRINT FUNCTION>
```

These objects designate their definitions, so `(docstring #'print)` works. Extending DRef and these operations is possible through **Defining Locative Types**. It is also possible to define new operations. For example, PAX makes `pax:document` extensible through `pax:document-object*`.

Finally, existing definitions can be queried with **definitions** and **dref-*apropos***:

```
(definitions 'dref-ext:arglist*)
==> (#<DREF ARGLIST* GENERIC-FUNCTION>
--> #<DREF ARGLIST* (METHOD (CLASS-DREF))>
--> #<DREF ARGLIST* (METHOD (COMPILER-MACRO-DREF))>
--> #<DREF ARGLIST* (METHOD (FUNCTION-DREF))>
--> #<DREF ARGLIST* (METHOD (LAMBDA-DREF))>
--> #<DREF ARGLIST* (METHOD (MACRO-DREF))>
--> #<DREF ARGLIST* (METHOD (METHOD-DREF))>
--> #<DREF ARGLIST* (METHOD (SETF-DREF))>
--> #<DREF ARGLIST* (METHOD (TYPE-DREF))> #<DREF ARGLIST* (METHOD (DREF))>
--> #<DREF ARGLIST* (METHOD (MGL-PAX::GO-DREF))>
--> #<DREF ARGLIST* (METHOD (T))>
```

```

--> #<DREF ARGLIST* (UNKNOWN
-->          (DECLAIM ARGLIST*
-->          FTYPE))>)

(dref-apospos 'locate-error :package :dref)
==> (#<DREF LOCATE-ERROR CONDITION> #<DREF LOCATE-ERROR FUNCTION>)

(dref-apospos "ate-err" :package :dref :external-only t)
==> (#<DREF LOCATE-ERROR CONDITION> #<DREF LOCATE-ERROR FUNCTION>)

```

### 3 References

After the [Introduction](#), here we get into the details. Of special interest are:

- The [xref](#) function to construct an arbitrary [reference](#) without any checking of validity.
- [locate](#) and [dref](#) to look up the [definition](#) of an object (e.g. `#'print`) or a [reference](#) (e.g. `(xref 'print 'function)`).
- [resolve](#) to find the first-class (non-[xref](#)) object the definition refers to, if any.

The [Basic Operations](#) ([arglist](#), [docstring](#), [source-location](#)) know how to deal with references (discussed in the [Extending DRef](#)).

- **[class]** `xref`

An `xref` (cross-reference) is a [reference](#). It may represent some kind of [definition](#) of its [name](#) in the context given by its [locative](#). The definition may not exist and the locative may even be [invalid](#). The subclass [dref](#) represents definitions that exist.

- **[function]** `xref` *name locative*

A shorthand for `(make-instance 'xref :name name :locative locative)` to create [xref](#) objects. It does no error checking: the [locative-type](#) of `locative` need not be defined, and the [locative-args](#) need not be valid. Use [locate](#) or the [dref](#) function to create [dref](#) objects.

- **[function]** `xref=` *xref1 xref2*

See if `xref1` and `xref2` have the same [xref-name](#) and [xref-locative](#) under `equal`. Comparing like this makes most sense for [drefs](#). However, two [xrefs](#) different under `xref=` may denote the same [drefs](#).

- **[class]** `dref` *xref*

`drefs` can be thought of as [definitions](#) that actually exist, although changes in the system can invalidate them (for example, a `dref` to a function definition can be invalidated by `fmakunbound`). `drefs` must be created with [locate](#) or the [dref](#) function.

Two `drefs` created in the same [dynamic environment](#) denote the same thing if and only if they are `xref=`.

- **[function] `locate`** *object &optional (errorp t)*

Return a [dref](#) representing the [definition](#) of object.

object must be a supported first-class object, a dref, or an [xref](#):

```
(locate #'print)
==> #<DREF PRINT FUNCTION>
```

```
(locate (locate #'print))
==> #<DREF PRINT FUNCTION>
```

```
(locate (xref 'print 'function))
==> #<DREF PRINT FUNCTION>
```

When object is a dref, it is simply returned.

Else, a `locate-error(0 1)` is signalled if object is an xref with an invalid [locative](#), or if no corresponding definition is found. If errorp is nil, then nil is returned instead.

```
(locate (xref 'no-such-function 'function))
.. debugger invoked on LOCATE-ERROR:
.. Could not locate NO-SUCH-FUNCTION FUNCTION.
.. NO-SUCH-FUNCTION does not name a function.
```

```
(locate (xref 'print '(function xxx)))
.. debugger invoked on LOCATE-ERROR:
.. Could not locate PRINT #'XXX.
.. Bad arguments (XXX) for locative FUNCTION with lambda list NIL.
```

```
(locate "xxx")
.. debugger invoked on LOCATE-ERROR:
.. Could not locate "xxx".
```

Use the [xref](#) function to construct an xref without error checking.

See [Extending locate](#).

- **[function] `dref`** *name locative &optional (errorp t)*

Shorthand for `(locate (xref name locative) errorp)`.

- **[function] `resolve`** *object &optional (errorp t)*

If object is an [xref](#), then return the first-class object associated with its definition if any. Return object if it's not an xref. Thus, the value returned is never an xref. The second return value is whether resolving succeeded.

```
(resolve (dref 'print 'function))
==> #<FUNCTION PRINT>
=> T
```

```
(resolve #'print)
==> #<FUNCTION PRINT>
```

```
=> T
```

If object is an xref, and the definition for it cannot be `located`, then `locate-error(0 1)` is signalled.

```
(resolve (xref 'undefined 'variable))  
.. debugger invoked on LOCATE-ERROR:  
.. Could not locate UNDEFINED VARIABLE.
```

If there is a definition, but there is no first-class object corresponding to it, then `resolve-error(0 1)` is signalled or `nil` is returned depending on `errorp`:

```
(resolve (dref '*print-length* 'variable))  
.. debugger invoked on RESOLVE-ERROR:  
.. Could not resolve *PRINT-LENGTH* VARIABLE.
```

```
(resolve (dref '*print-length* 'variable) nil)  
=> NIL  
=> NIL
```

`resolve` is a partial inverse of `locate`: if a `dref` is resolveable, then locating the object it resolves to recovers the `dref` equivalent to the original (`xref=` and of the same type but not `eq`).

Can be extended via `resolve*`.

- **[condition]** `locate-error` *error condition-context-mixin*

Signalled by `locate` when the definition cannot be found, and `errorp` is true.

- **[condition]** `resolve-error` *error condition-context-mixin*

Signalled by `resolve` when the object defined cannot be returned, and `errorp` is true.

### 3.1 Dissecting References

- **[reader]** `xref-name` *xref* (:name)

The `name` of the reference.

- **[reader]** `xref-locative` *xref* (:locative)

The `locative` of the reference.

The locative is normalized by replacing single-element lists with their only element:

```
(xref 'print 'function)  
==> #<XREF PRINT FUNCTION>
```

```
(xref 'print '(function))  
==> #<XREF PRINT FUNCTION>
```

- **[reader]** `dref-name` *dref*

The same as `xref-name`, but only works on `drefs`. Use it as a statement of intent.

- **[reader]** `dref-locative` *dref*

The same as `xref-locative`, but only works on `drefs`. Use it as a statement of intent.

- **[reader]** `dref-origin` *dref*

The object from which `locate` constructed this `dref`. `dref-origin` may have `presentation` arguments, which are not included in `locative-args` as is the case with the `initform` argument of the `variable` `locative`:

```
(dref '*standard-output*' '(variable "see-below"))  
==> #<DREF *STANDARD-OUTPUT* VARIABLE>
```

```
(dref-origin (dref '*standard-output*' '(variable "see-below")))  
==> #<XREF *STANDARD-OUTPUT* (VARIABLE "see-below")>
```

The `initform` argument overrides the global binding of `*standard-output*` when it's `pax:documented`:

```
(first-line  
 (pax:document (dref '*standard-output*' '(variable "see-below"))  
               :stream nil))  
=> "- [variable] *STANDARD-OUTPUT* \"see-below\""
```

- **[function]** `locative-type` *locative*

Return `locative type` of the `locative` `locative`. This is the first element of `locative` if it's a list. If it's a symbol, it's that symbol itself.

- **[function]** `locative-args` *locative*

Return the `rest` of `locative` `locative` if it's a list. If it's a symbol, then return `nil`. See `locative`.

The following convenience functions are compositions of `{locative-type, locative-args}` and `{xref-locative, dref-locative}`.

- **[function]** `xref-locative-type` *xref*
- **[function]** `xref-locative-args` *xref*
- **[function]** `dref-locative-type` *dref*
- **[function]** `dref-locative-args` *dref*

### 3.2 References Glossary

- **[glossary-term]** `reference`

A reference is a `name` plus a `locative`, and it identifies a possible definition. References are of class `xref`. When a reference is a `dref`, it may also be called a `definition`.

- **[glossary-term] definition**

A definition is a [reference](#) that identifies a concrete definition. Definitions are of class [dref](#). A definition [resolves](#) to the first-class object associated with the definition if such a thing exists, and [locate](#) on this object returns the canonical [dref](#) object that's unique under [xref=](#).

The kind of a definition is given by its [locative type](#). There is at most one definition for any given [name](#) and locative type. Equivalently, there can be no two definitions of the same [dref-name](#) and [dref-locative-type](#) but different [dref-locative-args](#).

- **[glossary-term] name**

Names are symbols, strings and nested lists of the previous, which name [functions](#), [types](#), [packages](#), etc. Together with [locatives](#), they form [references](#).

See [xref-name](#) and [dref-name](#).

- **[glossary-term] locative**

Locatives specify a *type* of definition such as [function](#) or [variable](#). Together with [names](#), they form [references](#).

In their compound form, locatives may have arguments (see [locative-args](#)) as in `(method (number))`. In fact, their atomic form is shorthand for the common no-argument case: that is, `function` is equivalent to `(function)`.

A locative is valid if it names an existing [locative type](#) and its `locative-args` match that type's lambda-list (see [define-locative-type](#)).

```
(arglist (dref 'method 'locative))  
=> (&REST QUALIFIERS-AND-SPECIALIZERS)  
=> :DESTRUCTURING
```

See [xref-locative](#) and [dref-locative](#).

- **[glossary-term] locative type**

The locative type is the part of a [locative](#) that identifies what kind of definition is being referred to. This is always a symbol.

Locative types are defined with [define-locative-type](#) or [define-pseudo-locative-type](#). See [Basic Locative Types](#) for the list locative types built into DRef, and PAX Locatives for those in PAX.

Also, see [locative-type](#), [xref-locative-type](#), [dref-locative-type](#), [Defining Locative Types](#).

- **[glossary-term] presentation**

[references](#) may have arguments (see [Defining Locative Types](#)) that do not affect the behaviour of [locate](#) and the [Basic Operations](#), but which may be used for other, "presentation" purposes. For example, the [variable](#) locative's `initform` argument is used for presentation by `pax:document`. Presentation arguments are available via [dref:dref-origin](#) but do not feature in [dref-locative](#) to ensure the uniqueness of the definition under [xref=](#).



## 4 dtypes

dtypes generalize [locative types](#) the same way Lisp types generalize [classes](#). A dtype is either

- a [locative type](#) such as [function](#), [type](#) and [clhs](#), or
- a full [locative](#) such as [\(method \(number\)\)](#) and [\(clhs section\)](#), or
- [nil](#) (the empty dtype) and [t](#) (that encompasses all [lisp-locative-types](#)), or
- named with [define-dtype](#) (such as [pseudo](#) and [top](#)), or
- a combination of the above with [and](#), [or](#) and [not](#), or
- a [member](#) form with [locateable](#) definitions, or
- a [satisfies](#) form with the name of a function that takes a single [definition](#) as its argument.

dtypes are used in [definitions](#) and [dref-apropos](#) to filter the set of definitions as in

```
(definitions 'print :dtype '(not unknown))  
==> (#<DREF PRINT (CLHS FUNCTION)> #<DREF PRINT FUNCTION>)
```

```
(dref-apropos "type specifier" :dtype 'pseudo)  
==> (#<DREF "1.4.4.6" #1=(CLHS SECTION)> #<DREF "1.4.4.6.1" #1#>  
--> #<DREF "1.4.4.6.2" #1#> #<DREF "1.4.4.6.3" #1#>  
--> #<DREF "1.4.4.6.4" #1#> #<DREF "4.2.3" #1#>  
--> #<DREF "atomic type specifier" #2=(CLHS GLOSSARY-TERM)>  
--> #<DREF "compound type specifier" #2#>  
--> #<DREF "derived type specifier" #2#> #<DREF "type specifier" #2#>)
```

- **[macro]** `define-dtype` *name lambda-list &body body*

Like [deftype](#), but it may expand into other dtypes.

The following example defines `method*` as the locative [method](#) without its direct locative subtypes.

```
(define-dtype method* () '(and method (not reader) (not writer)))
```

See [dtypep](#) for the semantics and also the locative [dtype](#).

- **[dtype]** `top`

This is the top of the dtype hierarchy, much like `t` for Lisp types. It expands to `(or t pseudo)`. While `t` matches every normal Lisp object and objectless definitions present in the running Lisp (see [lisp-locative-types](#)), `top` matches even pseudo definitions (see [pseudo-locative-types](#)).

- **[dtype]** `pseudo`

This is the union of all [pseudo-locative-types](#). It expands to `(or ,@(pseudo-locative-types))`.

- **[function]** `dtypep` *dref dtype*

See if dref is of dtype.

- *Atomic locatives*: If dtype is a *locative type*, then it matches definitions with that locative type and its locative subtypes.

Because *constant* is defined with *variable* among its *locative-supertypes*:

```
(dtypep (dref 'pi 'constant) 'variable)
=> T
```

```
(dtypep (dref 'number 'class) 'type)
=> T
```

It is an error if dtype is an *atom* but not a *locative type*. On the other hand, (the empty) argument list of atomic locatives is not checked even if having no arguments makes them *invalid*.

- *Compound locatives*: Locatives in their compound form are validated and must match exactly (under *equal*, as in *xref=*).

```
(defparameter *d* (dref 'dref* '(method (t t t))))
(defparameter *d2* (dref 'dref* '(method :around (t t t))))
(dtypep *d* 'method)
=> T
(dtypep *d* '(accessor))
.. debugger invoked on SIMPLE-ERROR:
.. Bad arguments NIL for locative ACCESSOR with lambda list
   ↪ (CLASS-NAME).
(dtypep *d* '(method (t t t)))
=> T
(dtypep *d2* '(method (t t t)))
=> NIL
```

- dtype may be constructed with *and*, *or* and *not* from Lisp types, locative types, full locatives and named dtypes:

```
(dtypep (dref 'locate-error 'condition) '(or condition class))
=> T
(dtypep (dref nil 'type) '(and type (not class)))
=> T
```

- For (member &rest objs), each of objs is *located* and dref is matched against them with *xref=*:

```
(dtypep (locate #'print) `(member ,#'print))
=> T
```

- For (satisfies pred), the predicate pred is funcalled with dref.
- dtype may be named by *define-dtype*:

```
(dtypep (locate #'car) 'top)
=> T
```

## 5 Listing Definitions

- **[function]** `definitions` *name &key (dtype t) (sort t)*

List all definitions of `name` that are of dtype as `drefs`.

Just as `(dref name locative)` returns the canonical definition, the `dref-names` of returned by definitions may be different from `name`:

```
(definitions "PAX")  
==> (#<DREF "MGL-PAX" PACKAGE>)
```

```
(definitions 'mgl-pax)  
==> (#<DREF "MGL-PAX" PACKAGE> #<DREF "mgl-pax" ASDF/SYSTEM:SYSTEM>)
```

Similarly, `dref-locative-type` may be more made more specific:

```
(definitions 'dref:locate-error :dtype 'type)  
==> (#<DREF LOCATE-ERROR CONDITION>)
```

If sort, the returned list is ordered according to `sort-references`.

Can be extended via `map-definitions-of-name`.

- **[function]** `dref-apropos` *name &key package external-only case-sensitive (dtype t) (sort t)*

Return a list of `drefs` corresponding to existing definitions that match the various arguments. First, `(dref-apropos nil)` lists all definitions in the running Lisp and maybe more (e.g. `mgl-pax:clhs`). Arguments specify how the list of definitions is filtered.

`dref-apropos` itself is similar to `cl:apropos-list`, but

- it finds `definitions` not `symbols`,
- it supports an extensible definition types, and
- filtering based on them.

PAX has a live browsing frontend.

Roughly speaking, when `name` or `package` is a `symbol`, it must match the whole `name` of the definition:

```
(dref-apropos 'method :package :dref :external-only t)  
==> (#<DREF METHOD CLASS> #<DREF METHOD LOCATIVE>)
```

On the other hand, when `name` or `package` is a `string`, they are matched as substrings to the definition's name `princ-to-stringed`:

```
(dref-apropos "method" :package :dref :external-only t)  
==> (#<DREF SETF-METHOD LOCATIVE> #<DREF METHOD CLASS>  
-> #<DREF METHOD LOCATIVE> #<DREF METHOD-COMBINATION CLASS>  
-> #<DREF METHOD-COMBINATION LOCATIVE>)
```

Definitions that are not of dtype (see `dtypep`) are filtered out:

```
(dref-apropos "method" :package :dref :external-only t :dtype 'class)
==> (#<DREF METHOD CLASS> #<DREF METHOD-COMBINATION CLASS>)
```

When package is :none, only non-symbol *names* are matched:

```
(dref-apropos "dref" :package :none)
==> (#<DREF "DREF" PACKAGE> #<DREF "DREF-EXT" PACKAGE>
--> #<DREF "DREF-TEST" PACKAGE> #<DREF "dref" ASDF/SYSTEM:SYSTEM>
--> #<DREF "dref/full" ASDF/SYSTEM:SYSTEM>
--> #<DREF "dref/test" ASDF/SYSTEM:SYSTEM>
--> #<DREF "dref/test-autoload" ASDF/SYSTEM:SYSTEM>)
```

The exact rules of filtering are as follows. Let *c* be the *name* of the candidate definition from the list of all definitions that we are matching against the arguments and denote its string representation (*princ-to-string c*) with *p*. Note that *princ-to-string* does not print the package of symbols. We say that two strings *match* if case-sensitive is nil and they are *equalp*, or case-sensitive is true and they are *equal*. case-sensitive affects *substring* comparisons too.

- If name is a symbol, then its *symbol-name* must *match* *p*.
- If name is a string, then it must be a *substring* of *p*.
- If package is :any, then *c* must be a symbol.
- If package is :none, then *c* must *not* be a symbol.
- If package is not nil, :any or :none, then *c* must be a symbol.
- If package is a *package*, it must be *eq* to the *symbol-package* of *c*.
- If package is a symbol other than nil, :any and :none, then its *symbol-name* must *match* the *package-name* or one of the *package-nicknames* of *symbol-package* of *c*.
- If package is a string, then it must be a *substring* of the *package-name* of *symbol-package* of *c*.
- If *external-only* and *c* is a symbol, then *c* must be external in a matching package.
- *dtype* matches candidate definition *d* if (*dtypep d dtype*).

If *sort*, the returned list is ordered according to *sort-references*.

Can be extended via MAP-REFERENCES-OF-TYPE and *map-definitions-of-name*.

- **[macro] `with-definitions-cached`** (*&key (dtype "top")*) *&body body*

Allow caching of definition lookups and *dtype* computations. This can provide a speedup when multiple calls are made to *definitions* or *dref-*apropos** within *body*.

Using this macro makes the promise that, during the *dynamic extent* of *body*, the following are unchanged:

- the set of *definitions* of *dtype*,

- the set of all [locative types](#),
- the set of all [dtypes](#).

Note that definitions *not* of dtype are allowed to be made or deleted.

- **[function]** `sort-references` *references &key key*

Order references in an implementation independent way.

- Non-symbol named references go first and are [sorted by the locative type](#) first, then by name and locative args.
- Symbol-based references go after and are sorted by name first, then [by the locative-type](#), finally by the locative args.

- **[function]** `sort-locative-types` *locative-types*

Sort locative-types in an implementation independent way. The order is alphabetical in [symbol-name](#), with the [package-name](#) acting as a tie breaker. If [unknown](#) is present among locative-types, it goes last.

- **[glossary-term]** `reverse definition order`

The lists of [locative types](#) returned by e.g. [locative-types](#), [lisp-locative-types](#), [pseudo-locative-types](#) and [locative-aliases](#) are in reverse order of the time of their definition. This order is not affected by redefinition, regardless of whether it's by [define-locative-type](#), [define-pseudo-locative-type](#), [define-symbol-locative-type](#) or [define-locative-alias](#).

- **[function]** `locative-types`

Return a list of non-[alias](#) locative types. This is the [union](#) of [lisp-locative-types](#) and [pseudo-locative-types](#), which is the set of constituents of the dtype [top](#).

This list is in [reverse definition order](#).

- **[function]** `lisp-locative-types`

Return the locative types that correspond to Lisp definitions, which typically have [source-location](#). These are defined with [define-locative-type](#) and [define-symbol-locative-type](#) and are the constituents of dtype [t](#).

This list is in [reverse definition order](#).

- **[function]** `pseudo-locative-types`

Return the locative types that correspond to non-Lisp definitions. These are the ones defined with [define-pseudo-locative-type](#) and are the constituents of dtype [pseudo](#).

This list is in [reverse definition order](#).

- **[function]** `locative-aliases`

Return the list of locatives aliases, defined with [define-locative-alias](#).

This list is in [reverse definition order](#).

## 6 Basic Operations

The following functions take a single argument, which may be a [dref](#), or an object denoting its own definition (see [locate](#)).

- **[function]** `arglist` *object*

Return the arglist of the definition of *object* or `nil` if the arglist cannot be determined.

The second return value indicates whether the arglist has been found. As the second return value, `:ordinary` indicates an [ordinary lambda list](#), `:macro` a [macro lambda list](#), `:deftype` a [deftype lambda list](#), and `:destructuring` a [destructuring lambda list](#). Other non-`nil` values are also allowed.

```
(arglist #'arglist)
=> (OBJECT)
=> :ORDINARY
```

```
(arglist (dref 'define-locative-type 'macro))
=> (LOCATIVE-TYPE-AND-LAMBDA-LIST LOCATIVE-SUPERTYPES &OPTIONAL
   DOCSTRING DREF-DEFCLASS-FORM)
=> :MACRO
```

```
(arglist (dref 'arglist* '(method (method-dref))))
=> ((DREF METHOD-DREF))
=> :SPECIALIZED
```

```
(arglist (dref 'method 'locative))
=> (&REST QUALIFIERS-AND-SPECIALIZERS)
=> :DESTRUCTURING
```

This function supports [macros](#), [compiler-macros](#), [setf](#) functions, [functions](#), [generic-functions](#), [methods](#), [types](#), [locatives](#). Note that `arglist` depends on the quality of `swank-backend:arglist`. With the exception of SBCL, which has perfect support, all Lisp implementations have minor omissions:

- [deftype](#) lambda lists on ABCL, AllegroCL, CLISP, CCL, CMUCL, ECL;
- default values in [macro](#) lambda lists on AllegroCL;
- various edge cases involving traced functions.

Can be extended via [arglist\\*](#).

- **[function]** `arglist-parameters` *arglist &optional (arglist-type :ordinary)*

A utility to extract the names of the parameters from `arglist` of `arglist-type`. See the function [arglist](#), for the possible values of `arglist-type`.

Note that if `arglist` is `nil`, then `arglist-type` may also be `nil`, in which case `nil` is returned.

```
(arglist #'source-location)
=> (OBJECT &KEY ERROR)
=> :ORDINARY
```

```
(multiple-value-call 'arglist-parameters (arglist #'source-location))
=> (OBJECT ERROR)
```

Can be extended via [arglist-parameters\\*](#).

- **[function]** `docstring` *object*

Return the docstring from the definition of *object*. As the second value, return the *\*package\** that was in effect when the docstring was installed or `nil` if it cannot be determined (this is used by `pax:document` when determining the Package and Readtable of docstrings). This function is similar in purpose to `cl:documentation`.

Note that some locative types such as [asdf:systems](#) and [declarations](#) have no docstrings, and some Lisp implementations do not record all docstrings. The following are known to be missing:

- [compiler-macro](#) docstrings on ABCL, AllegroCL, CCL, ECL;
- [method-combination](#) docstrings on ABCL, AllegroCL.

Can be extended via [docstring\\*](#).

- **[function]** `source-location` *object &key error*

Return the Swank source location for the *defining form* of *object*.

The returned Swank location object is to be accessed only through the [Source Locations](#) API or to be passed to e.g Slime's `slime-goto-source-location`.

If no source location was found,

- if *error* is `nil`, then return `nil`;
- if *error* is `:error`, then return a list of the form `(:error <error-message>)` suitable for `slime-goto-source-location`;
- if *error* is `t`, then signal an *error* condition with the same error message as in the previous case.

Note that the availability of source location information varies greatly across Lisp implementations.

With the `nil` [backend](#), `dref:source-location` loses precision beyond toplevel forms.

Can be extended via [source-location\\*](#).

## 7 Basic Locative Types

The following are the [locative types](#) supported out of the box. Like all locative types, they are named by symbols. When there is a CL type corresponding to the reference's locative type, the

references can be [resolved](#) to a unique object as is the case in

```
(resolve (dref 'print 'function))  
==> #<FUNCTION PRINT>  
=> T
```

Even if there is no such CL type, the [arglist](#), the [docstring](#), and the [source-location](#) of the defining form is usually recorded unless otherwise noted.

The basic locative types and their inheritance structure is loosely based on the doc - type argument of [cl:documentation](#).

## 7.1 Locatives for Variables

- **[locative] [variable](#)** *&optional initform*

- Direct locative subtypes: concept, glossary-term, section, [constant](#)

Refers to a global special variable. *initform*, or if not specified, the global value of the variable is to be used for [presentation](#).

```
(dref '*print-length* 'variable)  
==> #<DREF *PRINT-LENGTH* VARIABLE>
```

*variable* references do not [resolve](#).

- **[locative] [constant](#)** *&optional initform*

- Direct locative supertypes: [variable](#)

Refers to a constant variable defined with [defconstant](#). *initform*, or if not specified, the value of the constant is included in the documentation. The [constant](#) locative is like the [variable](#) locative, but it also checks that its object is [constantp](#).

*constant* references do not [resolve](#).

## 7.2 Locatives for Macros

- **[locative] [setf](#)**

- Direct locative subtypes: [setf-method](#), [setf-function](#)

Refers to a [setf expander](#) (see [defsetf](#) and [define-setf-expander](#)).

[Setf functions](#) (e.g. `(defun (setf name) ...)` or the same with [defgeneric](#)) are handled by the [setf-function](#), [setf-generic-function](#), and [setf-method](#) locatives.

*setf expander* references do not [resolve](#).

- **[locative] [macro](#)**

Refers to a global macro, typically defined with [defmacro](#), or to a [special operator](#).

*macro* references resolve to the [macro-function](#) of their name or signal `resolve-error(0 1)` if that's `nil`.



- [locative] **symbol-macro**

Refers to a global symbol macro, defined with `define-symbol-macro`. Note that since `define-symbol-macro` does not support docstrings, PAX defines methods on the documentation generic function specialized on `(doc-type (eql 'symbol-macro))`.

```
(define-symbol-macro my-mac 42)
(setf (documentation 'my-mac 'symbol-macro)
      "This is MY-MAC.")
(documentation 'my-mac 'symbol-macro)
=> "This is MY-MAC."
```

`symbol-macro` references do not `resolve`.

- [locative] **compiler-macro**

- Direct locative subtypes: `setf-compiler-macro`

Refers to a `compiler-macro-function`, typically defined with `define-compiler-macro`.

- [locative] **setf-compiler-macro**

- Direct locative supertypes: `compiler-macro`

Refers to a compiler macro with a `setf function name`.

`setf-compiler-macro` references do not `resolve`.

### 7.3 Locatives for Functions and Methods

- [locative] **function**

- Direct locative subtypes: `structure-accessor`, `setf-function`, `generic-function`

Refers to a global function, typically defined with `defun`. The `name` must be a `function name`. It is also allowed to reference `generic-functions` as functions:

```
(dref 'docstring 'function)
==> #<DREF DOCSTRING FUNCTION>
```

- [locative] **setf-function**

- Direct locative supertypes: `function`, `setf`
- Direct locative subtypes: `structure-accessor`, `setf-generic-function`

Refers to a global `function` with a `setf function name`.

```
(defun (setf ooh) ())
(locate #'(setf ooh))
==> #<DREF 00H SETF-FUNCTION>
(dref 'ooh 'setf-function)
==> #<DREF 00H SETF-FUNCTION>
(dref '(setf ooh) 'function)
==> #<DREF 00H SETF-FUNCTION>
```

- **[locative] generic-function**

- Direct locative supertypes: [function](#)
- Direct locative subtypes: [setf-generic-function](#)

Refers to a **generic-function**, typically defined with **defgeneric**. The **name** must be a **function name**.

- **[locative] setf-generic-function**

- Direct locative supertypes: [generic-function](#), [setf-function](#)

Refers to a global **generic-function** with a **setf function name**.

```
(defgeneric (setf oog) ())
(locate #'(setf oog))
==> #<DREF 00G SETF-GENERIC-FUNCTION>
(dref 'oog 'setf-function)
==> #<DREF 00G SETF-GENERIC-FUNCTION>
(dref '(setf oog) 'function)
==> #<DREF 00G SETF-GENERIC-FUNCTION>
```

- **[locative] method** *&rest qualifiers-and-specializers*

- Direct locative subtypes: [writer](#), [reader](#), [setf-method](#)

Refers to a **method**. **name** must be a **function name**. **method-qualifiers-and-specializers** has the form

```
(<QUALIFIER>* <SPECIALIZERS>)
```

For example, the method

```
(defgeneric foo-gf (x y z)
  (:method :around (x (y (eq1 'xxx)) (z string))
    (values x y z)))
```

can be referred to as

```
(dref 'foo-gf '(method :around (t (eq1 xxx) string)))
==> #<DREF F00-GF (METHOD :AROUND (T (EQL XXX) STRING))>
```

**method** is not exportable-locative-type-p.

- **[locative] setf-method** *&rest method-qualifiers-and-specializers*

- Direct locative supertypes: [method](#), [setf](#)
- Direct locative subtypes: [accessor](#)

Refers to a **method** of a **setf-generic-function**.

```
(defgeneric (setf oog) (v)
  (:method ((v string))))
(locate (find-method #'(setf oog) ()) (list (find-class 'string))))
==> #<DREF 00G (SETF-METHOD (STRING))>
```

```
(dref 'oog '(setf-method (string)))
==> #<DREF 00G (SETF-METHOD (STRING))>
(dref '(setf oog) '(method (string)))
==> #<DREF 00G (SETF-METHOD (STRING))>
```

- **[locative]** **method-combination**

Refers to a **method-combination**, defined with **define-method-combination**.

**method-combination** references do not **resolve**.

- **[locative]** **reader** *class-name*

- Direct locative supertypes: **method**

- Direct locative subtypes: **accessor**

Refers to a **:reader** method in a **defclass**:

```
(defclass foo ()
  ((xxx :reader foo-xxx)))

(dref 'foo-xxx '(reader foo))
==> #<DREF F00-XXX (READER F00)>
```

- **[locative]** **writer** *class-name*

- Direct locative supertypes: **method**

- Direct locative subtypes: **accessor**

Like **accessor**, but refers to a **:writer** method in a **defclass**.

- **[locative]** **accessor** *class-name*

- Direct locative supertypes: **reader**, **writer**, **setf-method**

Refers to an **:accessor** in a **defclass**.

An **:accessor** in **defclass** creates a reader and a writer method. Somewhat arbitrarily, **accessor** references **resolve** to the writer method but can be **located** with either.

- **[function]** **accessor-slot-definition** *dref*

Return the **SLOT-DEFINITION** object corresponding to **dref**, which may denote a reader, a writer or an accessor.

- **[locative]** **structure-accessor** *&optional structure-class-name*

- Direct locative supertypes: **setf-function**, **function**

Refers to an accessor function generated by **defstruct** or **defstruct\***. A **locate-error** condition is signalled if the wrong **structure-class-name** is provided.

Note that there is no portable way to detect structure accessors, and on some platforms, **(locate #'my-accessor)**, **definitions** and **dref-a-propos** will return **function** references instead. On such platforms, **structure-accessor** references do not **resolve**.

- **[macro] `defstruct*`** *name-and-options &rest slot-descriptions*

Like `defstruct`, but support `:documentation` among slot options. The documentation is attached to the slot's `structure-accessor`. Example:

```
(defstruct* my-struct
  (my-slot nil :documentation "docstring"))
```

In addition to the normal `defstruct` processing, the above also does the moral equivalent of

```
(setf (documentation 'my-struct-my-slot 'function) "docstring")
```

## 7.4 Locatives for Types and Declarations

- **[locative] `type`**
  - Direct locative subtypes: `class`

This locative can refer to `types` and `classes` as well as `conditions`, simply put, to things defined by `deftype`, `defclass` and `define-condition`.

```
(deftype my-type () t)
(dref 'my-type 'type)
==> #<DREF MY-TYPE TYPE>
```

```
(dref 'xref 'type)
==> #<DREF XREF CLASS>
```

```
(dref 'locate-error 'type)
==> #<DREF LOCATE-ERROR CONDITION>
```

type references do not `resolve`.

- **[locative] `class`**
  - Direct locative supertypes: `type`
  - Direct locative subtypes: `condition`, `structure`

Naturally, `class` is the locative type for `classes`.

`arglist` returns the `compound type specifier` syntax associated with `system classes`. E.g. for the `integer` class:

```
(arglist (find-class 'integer))
=> (&OPTIONAL LOWER-LIMIT UPPER-LIMIT)
=> :DEFTYPE
```

- **[locative] `structure`**
  - Direct locative supertypes: `class`

Refers to a `structure-class`, typically defined with `defstruct`.

Also, see [defstruct\\*](#).

- **[locative] declaration**

Refers to a declaration, used in [declare](#), [declaim](#) and [proclaim](#).

User code may also define new declarations with CLTL2 functionality, but their [arglist](#) is always `nil`. On implementations that support it, [docstring](#) returns (documentation name 'declaration).

```
(cl-environments:define-declaration my-decl (&rest things)
  (values :declare (cons 'foo things)))
```

declaration references do not [resolve](#).

Also, [source-location](#) on declarations currently only works on SBCL.

## 7.5 Locatives for the Condition System

- **[locative] condition**

- Direct locative supertypes: [class](#)

Although `condition` is not [subtypep](#) of `class`, actual condition objects are commonly instances of a condition class that is a CLOS class. HyperSpec [ISSUE:CLOS-CONDITIONS](#) and [ISSUE:CLOS-CONDITIONS-AGAIN](#) provide the relevant history.

Whenever a `class` denotes a condition, its [dref-locative-type](#) will be `condition`:

```
(dref 'locate-error 'class)
==> #<DREF LOCATE-ERROR CONDITION>
```

- **[locative] restart**

A locative to refer to the definition of a restart defined by [define-restart](#).

- **[macro] define-restart** *symbol lambda-list &body docstring*

Associate a definition with the name of a restart, which must be a symbol. `lambda-list` should be what calls like `(invoke-restart '<symbol> ...)` must conform to, but this is not enforced.

PAX "defines" standard CL restarts such as [use-value](#) with `define-restart`:

```
(dref 'use-value 'restart)
==> #<DREF USE-VALUE RESTART>
(assert (docstring *))
(assert (source-location **))
```

Note that while there is a `cl:restart` class, its *instances* have no `docstring` or `source location`.

## 7.6 Locatives for Packages and Readtables

- **[locative] asdf/system:system**

Refers to an already loaded `asdf:system` (those in `asdf:registered-systems`). The `name` may be anything `asdf:find-system` supports.

`asdf:system` is not `exportable-locative-type-p`.

- **[locative] package**

Refers to a `package`, defined by `defpackage` or `make-package`. The `name` may be anything `find-package` supports.

`package` is not `exportable-locative-type-p`.

- **[locative] readtable**

Refers to a named `readtable` defined with `named-readtables:defreadtable`, which associates a global name and a docstring with the readtable object. The `name` may be anything `find-readtable` supports.

`readtable` references `resolve` to `find-readtable` on their `name`.

## 7.7 Locatives for Unknown Definitions

- **[locative] unknown** *dspec*

This locative type allows PAX to work in a limited way with definition types it doesn't know. `unknown` definitions come from `definitions`, which uses `swank/backend:find-definitions`. The following examples show PAX stuffing the Swank `dspec` (`:define-alien-type double-float`) into an `unknown` locative on SBCL.

```
(definitions 'double-float)
==> (#<DREF DOUBLE-FLOAT CLASS>
--> #<DREF DOUBLE-FLOAT (UNKNOWN (:DEFINE-ALIEN-TYPE DOUBLE-FLOAT))>)
```

```
(dref 'double-float '(unknown (:define-alien-type double-float)))
==> #<DREF DOUBLE-FLOAT (UNKNOWN (:DEFINE-ALIEN-TYPE DOUBLE-FLOAT))>
```

`arglist` and `docstring` return `nil` for unknowns, but `source-location` works.

## 7.8 Locatives for DRef Constructs

- **[locative] dtype**

- Direct locative subtypes: `locative`

Locative for `dtypes` defined with `define-dtype` and `locative` types. `dtype` is to `locative` as `type` is to `class`.

The `top` of the `dtype` hierarchy:

```
(dref 'top 'dtype)
==> #<DREF TOP DTYPE>
```

This very definition:

```
(dref 'dtype 'locative)
==> #<DREF DTYPE LOCATIVE>
```

- **[locative]** **locative**

- Direct locative supertypes: [dtype](#)

This is the locative for [locative](#) types defined with [define-locative-type](#), [define-pseudo-locative-type](#) and [define-locative-alias](#).

```
(first-line (source-location-snippet
              (source-location (dref 'macro 'locative))))
=> "(define-locative-type macro ())"
```

- **[locative]** **lambda** *&key arglist arglist-type docstring docstring-package file file-position snippet &allow-other-keys*

A [pseudo locative type](#) that carries its [arglist](#), [docstring](#) and [source-location](#) in the locative itself. See [make-source-location](#) for the description of [file](#), [file-position](#), and [snippet](#). [lambda](#) references do not [resolve](#). The [name](#) must be `nil`.

```
(arglist (dref nil '(lambda :arglist ((x y) z)
                        :arglist-type :macro)))
=> ((X Y) Z)
=> :MACRO
```

```
(docstring (dref nil '(lambda :docstring "xxx"
                              :docstring-package :dref)))
=> "xxx"
==> #<PACKAGE "DREF">
```

```
(source-location-file
 (source-location (dref nil '(lambda :file "xxx.el"))))
=> "xxx.el"
```

Also, see the `pax:include` locative.

## 8 Backends

On SBCL, the `swank asdf:system` is not a dependency of [dref/full](#) because `DRef` is able to function without it. However, when `Swank` is available in the Lisp image, `DRef` can use it to provide more precise [source-locations](#).

On other Lisps, `DRef` currently gets [arglists](#) and [source-locations](#) via `Swank` and `swank` is an ASDF dependency of `dref/full`.

- **[function]** **backend-available-p** *backend*

Check if `backend` is currently available in the running Lisp. `backend` may be `:swank` or `nil`. Currently, on SBCL, both are available; on other Lisps, only `:swank` is.

- **[function]** **backend**

Return the backend being used. This is either `:swank` or `nil`. The default backend is `:swank` if it is available when first loading `dref` into the image, else it is `nil`.

Note that on Lisps other than SBCL, this is currently always `:swank`, which is available due to the ASDF dependency of `dref/full` on the `swank asdf:system`.

- **[setf] `backend`** *backend*

setting `backend` to a backend that is not `backend-available-p` is an error.

- **[macro] `with-backend`** (*backend*) &body *body*

Like `setf backend`, but only change backend during the dynamic extent of *body*.

## 9 Extending DRef

### 9.1 Extension Tutorial

Let's see how to tell DRef about new kinds of definitions through the example of the implementation of the `class` locative. Note that this is a verbatim `pax:include` of the sources. Please ignore any internal machinery. The first step is to define the `locative type`:

```
(define-locative-type class (type)
  "Naturally, CLASS is the locative type for [CLASS][class]es.

  ARGLIST returns the [compound type specifier][clhs] syntax
  associated with [system class][clhs]es. E.g. for the [INTEGER][clhs]
  class:

  ```cl-transcript (:dynenv dref-std-env)
  (arglist (find-class 'integer))
  => (&OPTIONAL LOWER-LIMIT UPPER-LIMIT)
  => :DEFTYPE
  ```")
```

Then, we make it possible to look up `class` definitions:

```
(define-locator class ((class class))
  (make-instance 'class-dref :name (class-name class) :locative 'class))

(define-lookup class (symbol locative-args)
  (unless (and (symbolp symbol)
               (find-class symbol nil))
    (locate-error "~S does not name a class." symbol))
  (make-instance 'class-dref :name symbol :locative 'class))
```

`define-locator` makes `(locate (find-class 'dref))` work, while `define-lookup` is for `(dref 'dref 'class)`. Naturally, for locative types that do not define first-class objects, the first method cannot be defined.

Finally, we define a `resolve*` method to recover the `class` object from a `class-dref`. We also specialize `docstring*` and `source-location*`:



```

(defmethod resolve* ((dref class-dref))
  (find-class (dref-name dref)))

(defmethod docstring* ((class class))
  (documentation* class t))

(defmethod source-location* ((dref class-dref))
  #+sbcl
  (sb-one-source-location (dref-name dref) :class)
  #-sbcl
  (swank-source-location* (resolve dref) (dref-name dref) 'class))

(defmethod arglist* ((dref class-dref))
  (clhs-type-specifier-arglist (dref-name dref)))

```

We took advantage of having just made the class locative type being `resolveable`, by specializing `docstring*` on the `class` class. `source-location*` was specialized on `class-dref` to demonstrate how this can be done for non-`resolveable` locative types.

Classes have no arglist, so no `arglist*` method is needed. In the following, we describe the pieces in detail.

## 9.2 Locative Type Hierarchy

`Locative types` form their own hierarchy, that is only superficially similar to the Lisp `class` hierarchy. The hierarchies of `lisp-locative-types` and `pseudo-locative-types` are distinct. That is, the `dref-class` of a Lisp locative type must not be a subclass of a `pseudo` one, and vice versa. This is enforced by `define-locative-type` and `define-pseudo-locative-type`.

- **[function]** `dref-class` *locative-type*

Return the name of the `class` used to represent `definitions` with `locative-type`. This is always a subclass of `dref`. Returns `nil` if `locative-type` is not a valid locative type.

Note that the actual `type-of` a `dref` is mostly intended for `Extending DRef`. Hence, it is hidden when a `dref` is printed:

```

(dref 'print 'function)
==> #<DREF PRINT FUNCTION>
(type-of *)
=> FUNCTION-DREF

```

Due to `Canonicalization`, the actual type may be a proper subtype of `dref-class`:

```

(dref 'documentation 'function)
==> #<DREF DOCUMENTATION GENERIC-FUNCTION>
(type-of *)
=> GENERIC-FUNCTION-DREF
(subtypep 'generic-function-dref 'function-dref)
=> T
=> T

```

- **[function]** `locative-type-direct-supers` *locative-type*

List the [locative types](#) whose [dref-classes](#) are direct superclasses of the `dref-class` of `locative-type`. These can be considered supertypes of `locative-type` in the sense of [dtypep](#).

This is ordered as in the corresponding definition.

- **[function]** `locative-type-direct-subs` *locative-type*

List the [locative types](#) whose [dref-classes](#) are direct subclasses of the `dref-class` of `locative-type`. These can be considered subtypes of `locative-type` in the sense of [dtypep](#).

This list is in [reverse definition order](#).

- **[function]** `locative-subtype-p` *locative-type-1 locative-type-2*

Check if `locative-type-1` is in the transitive closure of `locative-type-2` via [locative-type-direct-subs](#). It is an error if `locative-type-1` or `locative-type-2` is not a valid locative type.

### 9.3 Defining Locative Types

- **[macro]** `define-locative-type` *locative-type-and-lambda-list locative-supertypes &optional docstring dref-defclass-form*

Declare `locative-type` as a [locative](#), which is the first step in [Extending DRef](#).

- *Simple example*

To define a locative type called `dummy` that takes no arguments and is not a locative subtype of any other locative type:

```
(define-locative-type dummy ()
  "Dummy docstring.")
```

With this definition, only the locatives `dummy` and its equivalent form `(dummy)` are valid. The above defines a `dref(0 1)` subclass called `dummy-dref` in the current package. All definitions with locative type `dummy` and its locatives subtypes must be instances of `dummy-dref`.

`(locate 'dummy 'locative)` refers to this definition. That is, [arglist](#), [docstring](#) and [source-location](#) all work on it.

- *Complex example*

`dummy` may have arguments `x` and `y` and inherit from locative types `l1` and `l2`:

```
(define-locative-type (dummy x &key y) (l1 l2)
  "Dummy docstring."
  (defclass dummy-dref ()
    ((xxx :initform nil :accessor dummy-xxx))))
```

One may change name of dummy-dref, specify superclasses and add slots as with `defclass`. Behind the scenes, the dref classes of `l1` and `l2` are added automatically to the list of superclasses.

Arguments:

- The general form of `locative-type-and-lambda-list` is `(locative-type &rest lambda-list)`, where `locative-type` is a **symbol**, and `lambda-list` is a **destructuring lambda list**. The `locative-args` of drefs with `locative type` `locative-type` (the argument given to this macro) always conform to this lambda list. See `check-locative-args`.

If `locative-type-and-lambda-list` is a single symbol, then that's interpreted as `locative-type`, and `lambda-list` is `nil`.

- `locative-supertypes` is a list of **locative types** whose **dref-classes** are prepended to the list of superclasses this definition.

Locative types defined with `define-locative-type` can be listed with `lisp-locative-types`.

- **[macro]** `define-pseudo-locative-type` *locative-type-and-lambda-list locative-supertypes &optional docstring dref-defclass-form*

Like `define-locative-type`, but declare that `locative-type` does not correspond to definitions in the running Lisp. Definitions with pseudo locatives are of dtype `pseudo` and are not listed by default by `definitions`.

Locative types defined with `define-pseudo-locative-type` can be listed with `pseudo-locative-types`.

- **[macro]** `define-locative-alias` *alias locative-type &body docstring*

Define `alias` that can be substituted for `locative-type` (both **symbols**) for the purposes of `locateing`. `locative-type` must exist (i.e. be among `locative-types`). For example, let's define `object` as an alias of the `class` `locative`:

```
(define-locative-alias object class)
```

Then, `locateing` with `object` will find the `class`:

```
(dref 'xref 'object)
==> #<DREF XREF CLASS>
```

The `locative-args` of `object` (none in the above) are passed on to `class`.

```
(arglist (dref 'object 'locative))
=> (&REST ARGS)
=> :DESTRUCTURING
```

Note that **locative-aliases** are not `locative-types` and are not valid dtypes.

Also, see `Locative Aliases` in PAX.

### 9.3.1 Symbol Locatives

Let's see how the opaque `define-symbol-locative-type` and the obscure `define-definer-for-symbol-locative-type` macros work together to simplify the common task of associating definition with a symbol in a certain context.

- **[macro] `define-symbol-locative-type`** *locative-type-and-lambda-list locative-supertypes &optional docstring dref-class-def*

Similar to `define-locative-type`, but it assumes that all things `locateable` with `locative-type` are going to be symbols defined with a definer defined with `define-definer-for-symbol-locative-type`. Symbol locatives are for attaching a definition (along with arglist, documentation and source location) to a symbol in a particular context. An example will make everything clear:

```
(define-symbol-locative-type direction ()
  "A direction is a symbol.")

(define-definer-for-symbol-locative-type define-direction direction
  "With DEFINE-DIRECTION, one can document what a symbol means when
  interpreted as a DIRECTION.")

(define-direction up ()
  "UP is equivalent to a coordinate delta of (0, -1).")
```

After all this, `(dref 'up 'direction)` refers to the `define-direction` form above.

The `dref-class` of the defined locative type inherits from `symbol-locative-dref`, which may be used for specializing when implementing new operations.

- **[macro] `define-definer-for-symbol-locative-type`** *name locative-type &body docstring*

Define a macro with `name` that can be used to attach a lambda list, documentation, and source location to a symbol in the context of `locative-type`. The defined macro's arglist is `(symbol lambda-list &optional docstring)`. `locative-type` is assumed to have been defined with `define-symbol-locative-type`.

## 9.4 Extending `locate`

Internally, `locate` finds an initial `dref` of its object argument with a `lookup` or with a `locator`. This initial `dref` is then canonicalized with a series of `casts`. In more detail, the process is as follows.

- If the object argument of `locate` is a `dref`, then it is returned without processing.

Else, `locate` first needs to find the initial definition.

### 9.4.1 Initial Definition

`locate` can find the initial definition in one of two ways:

- *With direct lookup*

If object is an `xref(0 1)`, then the `lookup` for (`xref-locative-type` object) is invoked. For an `xref` with the locative (`method (number)`), this would be the lookup defined as

```
(define-lookup method (name locative-args) ...)
```

- *With locator search*

Else, object is a normal Lisp object, such as a `method` object from `find-method`. The first of `lisp-locative-types` whose `locator` succeeds provides the initial definition, which may be defined like this:

```
(define-locator method ((obj method)) ...)
```

This is a locator that returns definitions with the `method` locative type and takes an argument named `obj` of class `method` (which is like a `specializer` in `defmethod`).

- `lisp-locative-types` are tried one by one in the order specified there.
- For a given locative type, if there are multiple locators, standard CLOS method selection applies.

## 9.4.2 Canonicalization

The initial definition thus found is then canonicalized so that there is a unique `definition` under `xref=`:

```
(locate #'arglist*)  
==> #<DREF ARGLIST* GENERIC-FUNCTION>  
(dref 'arglist* 'function)  
==> #<DREF ARGLIST* GENERIC-FUNCTION>  
(dref 'arglist* 'generic-function)  
==> #<DREF ARGLIST* GENERIC-FUNCTION>
```

Canonicalization is performed by recursively attempting to `downcast` the current definition to one of its `locative-type-direct-subs` in a depth-first manner, backtracking if a cast fails.

**Default Downcast** By default, downcasting to `direct locative subtypes` is performed by looking up the definition where the locative type is replaced with its sub while the name and the locative args remain the same.

**Cast Name Change** `Casts` must be careful about changing `dref-name`.

Their `dref` argument and the `dref` returned must have the same `dref-name` (under `equal`, see `xref=`) or it must be possible to upcast the returned value to the `dref` argument's `dref-locative-type`.

- *Implementation note*

The purpose of this rule is to allow `dtypep` answer this correctly:

```
(defclass foo ()  
  ((a :accessor foo-a)))
```

```

(dref '(setf foo-a) '(method (t foo)))
==> #<DREF F00-A (ACCESSOR F00)>
(dtypep * '(method (t foo)))
=> T
;; Internally, DTYPEP upcasts #<DREF F00-A (ACCESSOR F00)>
;; and checks that the locative args of the resulting
;; definition match those in (METHOD (T F00)).
(locate* ** 'method)
==> #<DREF (SETF F00-A) (METHOD (T F00))>

```

For even more background, also note that if the name remains the same but locative args change, then `dtypep` can simply check with `dref` if there is a definition of the name with the given locative:

```

(defclass foo ()
  ((r :reader foo-r)))
(dref 'foo-r '(reader foo))
==> #<DREF F00-R (READER F00)>
(dtypep * '(method (foo)))
=> T
;; Behind the scenes, DTYPEP does this:
(xref= ** (dref 'foo-r '(method (foo))))
=> T

```

### 9.4.3 Defining Lookups, Locators and Casts

As we have seen, the [Initial Definition](#) is provided either by a lookup or a locator, then [Canonicalization](#) works with casts. Here, we look at how to define these.

*Implementation note:* All three are currently implemented as methods of generic functions with [eql specializers](#) for the locative type, which may easily prove to be problematic down the road. To make future changes easier, the generic function and the methods are hidden behind e.g. the `define-lookup` and `call-lookup` macros.

- **[variable]** `*check-locate*` *nil*

Enable runtime verification of invariants during `locate` calls. This carries a performance penalty and is intended for testing and debugging.

In particular, enforce the rule of [Cast Name Change](#) and that [lookups](#), [locators](#) and [casts](#) obey the following:

- The value returned must be either `nil` or a `dref(0 1)`. Alternatively, `locate-error(0 1)` may be signalled.
- If a `dref` is returned, then its `dref-locative-type` must be `locative-type`, and its class must be the `dref-class` of `locative-type`.
- `locative-args` must be congruent with the destructuring lambda list in the definition of `locative-type`.

- **[macro]** `define-lookup` *locative-type (name locative-args) &body body*

Define a method of looking up [definitions](#) of `locative-type` with the given `locative-args`. Lookups are invoked by `locate` when its object argument is an `xref(0 1)` with `locative-type` but it is not a `dref(0 1)`, as in the case of `(dref 'print 'function)`. When called, the variables `name` and `locative-args` are bound to `xref-name` and `xref-locative-args` of the `xref`. `locative-args` is validated with `check-locative-args` before body is evaluated.

```
(define-lookup variable (name locative-args)
  (unless (special-variable-name-p name)
    (locate-error))
  (make-instance 'variable-dref :name name :locative 'variable))
```

- `locative-type` is a valid [locative type](#).
- `name` and `locative-args` are both [symbols](#).

The above are enforced at macro-expansion time.

- body must follow the rules in `*check-locate*`.

- **[macro]** `call-lookup` *name locative-type locative-args*

Call the [lookup](#) for `locative-type` with `name` and `locative-args`.

- **[macro]** `define-locator` *locative-type ((object class)) &body body*

Define a method of finding the [definition](#) with `locative-type` of instances of `class`. When a locator's body is evaluated, `object` is bound to such an instance.

```
(define-locator class ((class class))
  (make-instance 'class-dref :name (class-name class) :locative 'class))
```

- `locative-type` is one of [lisp-locative-types](#). This is because [pseudo-locative-types](#) never [resolve](#) to first-class objects.
- `object` is a [symbol](#).
- `class` names a `class` that is not a subtype of `xref`. For how to convert definitions from one locative type to another, see [define-cast](#).

The above are enforced at macro-expansion time.

- body must follow the rules in `*check-locate*`.

In contrast to when the [Initial Definition](#) is created from an `xref` (see [define-lookup](#)), here `locative-args` are determined from `object`.

- **[macro]** `call-locator` *object locative-type*

Call the [locator](#) for `locative-type` with `object`.

- **[macro]** `define-cast` *locative-type ((dref dref-class)) &body body*

Define a method of converting a [definition](#) to another with `locative-type`. When a cast's body is evaluated, `dref` is bound to an instance `dref-class`, which denotes a valid but potentially [non-canonical](#) definition.

Note that the [Default Downcast](#) often suffices, and defining a cast is only necessary if the [name](#) or the [locative args](#) change:

```
(define-cast accessor ((dref reader-dref))
  (let ((name (dref-name dref))
        (class (second (dref-locative dref))))
    (when (ignore-errors (find-accessor-slot-definition name class))
      (make-instance 'accessor-dref :name name
                        :locative `(accessor ,class))))))
```

- [locative-type](#) is a valid [locative type](#).
  - If [locative-type](#) is one of [pseudo-locative-types](#), then [dref-class](#) must be of another pseudo [locative type](#).
  - [dref-class](#) is either a direct *downcast* or a potentially non-direct *upcast*.
    - \* *Downcast*: In this case, [locative-type](#) is one of [locative-type-direct-sub](#)s of ([dref-class-to-locative-type](#) [dref-class](#)).
- Downcasting to non-direct subtypes is done in multiple steps. Consequently, the body of a downcast can rely on ([class-of](#) [dref](#)) being [class](#), not any subclass thereof.
- \* *Upcast*: [locative-type](#) is different but reachable from ([dref-class-to-locative-type](#) [dref-class](#)) by repeatedly choosing one of [locative-type-direct-supers](#). Upcasting to non-direct supertypes is done in one step.

The above are enforced at macro-expansion time.

- body must follow the rules in [\\*check-locate\\*](#), including those in [Cast Name Change](#).
- **[macro]** `call-cast` *locative-type dref*  
Call the [cast](#) to [locative-type](#) with [dref](#).
  - **[function]** `locate-error` *&optional format-control &rest format-args*  
Call this function to signal a [locate-error](#) condition from the [dynamic extent](#) of a [locate](#) call, that is, from the bodys of [define-lookup](#), [define-locator](#) and [define-cast](#). It is an error to call `locate-error` elsewhere.  
  
[format-control](#), if non-`nil`, is a [format control](#) for which [format-args](#) are suitable.
  - **[macro]** `check-locative-args` *locative-type locative-args*  
Signal a [locate-error](#) condition if [locative-args](#) do not match the `lambda-list` argument of [locative-type](#) (not evaluated).

## 9.5 Extending Everything Else

- **[generic-function]** `resolve*` *dref*



Return the object defined by the definition `dref` refers to. Signal a `resolve-error` condition by calling the `resolve-error` function if the lookup fails.

To keep `resolve` a partial inverse of `locate`, `define-locator` may be necessary for resolveable definitions. This function is for extending `resolve`. Do not call it directly.

It is an error for methods of this generic function to return an `xref`.

- **[function]** `resolve-error` *&rest format-and-args*

Call this function to signal a `resolve-error` condition from the `dynamic extent` of a `resolve*` method. It is an error to call `resolve-error` elsewhere.

`format-and-args`, if non-`nil`, is a format string and arguments suitable for `format`.

- **[generic-function]** `map-definitions-of-name` *fn name locative-type*

Call `fn` with `drefs` which can be `located` with an `xref(0 1)` with `name`, `locative-type` and some `locative-args`. The strange wording here is because there may be multiple ways (and thus `xrefs`) that refer to the same definition.

For most `locative` types, there is at most one such definition, but for `method`, for example, there may be many. The default method simply does `(dref name locative-type nil)` and calls `fn` with result if `dref` succeeds.

`fn` is not called with the same (under `xref=`) definition multiple times.

- **[generic-function]** `map-definitions-of-type` *fn locative-type*

Call `fn` with `drefs` that can be `located` with an `xref(0 1)` with `locative-type` with some `name` and `locative-args`.

- Return `nil` if all possibilities have been exhausted.
- Return `true` if there may be definitions left unmapped. In this case, the caller is responsible for trying all interned symbols with `map-definitions-of-name`. The default method simply returns `true`.

`fn` may be called with `drefs` that are `xref=` and differ only in their `dref-origin`.

- **[generic-function]** `arglist*` *object*

To extend `arglist`, specialize `object` on a normal Lisp type or on a subclass of `dref`.

`arglist` first calls `arglist*` with its `object` argument. If that doesn't work (i.e. the second value returned is `nil`), then it calls `arglist*` with `object` either `resolved` (if it's a `dref`) or `located` (if it's not a `dref`).

- The default method returns `nil, nil`.
- There is also a method specialized on `drefs`, that looks up the `definition-property` called `arglist` and returns its value with `values-list`. Thus, an `arglist` and its kind can be specified with something like

```
(setf (definition-property xref 'arglist)
      (list arglist :destructuring))
```

This function is for extension only. Do not call it directly.

- **[generic-function]** `arglist-parameters*` *arglist arglist-type*

To extend `arglist-parameters`, *eql-specialize* `arglist-type` on, say, `:BOA` (see *BOA lambda lists*). This must be done if `arglist` is extended to return an unsupported value.

This function is for extension only. Do not call it directly.

- **[generic-function]** `docstring*` *object*

To extend `docstring`, specialize `object` on a normal Lisp type or on a subclass of `dref`.

`docstring` first calls `docstring*` with its `object` argument. If that doesn't work (i.e. `nil` is returned), then it calls `docstring*` with `object` either `resolved` (if it's a `dref`) or `located` (if it's not a `dref`).

- The default method returns `nil`.
- A three-stage logic is implemented by an unspecialized `:around` method for `docstrings` and `packages`.
  - \* First, the primary method is called. If it returns a non-`nil` `docstring` and `package`, then these are returned by `docstring*`.
  - \* *Definition default:* The `definition-property` with indicator `docstring` of the `located` definition of `object` provides defaults for the `docstring` and/or the `package`, whichever are `nil`. These can be set for example as

```
(setf (definition-property xref 'docstring)
      (list docstring *package*))
```

Note that the `docstring` and the `package` or both may be `nil`.

- \* *Package-wide default:* If the `dref-name` of the `located` definition is a symbol, then an irregular `definition-property` provides further defaults. To default the `docstring` `package` of all names of `package x` to `package y`, you could write

```
(setf (definition-property `(:package ,(find-package :x)) 'docstring)
      (list "FIXME: Missing documentation." (find-package :y)))
```

See `Package` and `Readtable`.

This function is for extension only. Do not call it directly.

- **[generic-function]** `source-location*` *object*

To extend `source-location`, specialize `object` on a normal Lisp type or on a subclass of `dref`.

`source-location` first calls `source-location*` with its `object` argument. If that doesn't work (i.e. `nil` or `(:error <message>)` is returned), then it calls `source-location*` with `object` either `resolved` (if it's a `dref`) or `located` (if it's not a `dref`).

`source-location` returns the last of the `(:error <message>)`s encountered or a generic error message if only `nil`s were returned.

- The default method returns `nil`.
- There is also a method specialized on `drefs`, that looks up the `definition-property` called `source-location`. If present, it must be a function of no arguments that returns a source location or `nil`. Typically, this is set up in the defining macro like this:

```
(setf (definition-property xref 'source-location)
      (this-source-location))
```

This function is for extension only. Do not call it directly.

- **[macro]** `nth-value-or-with-obj-or-def` (*obj nth-value*) &body *body*

Evaluate *body*. If its *nth-value* is `nil`, evaluate it again with *obj* bound to a `resolved` object (if *obj* was a definition) or a definition (if *obj* was not a definition). This is intended for implementing new operations with the fallback mechanism of `arglist`, `docstring` and `source-location*`.

### 9.5.1 Definition Properties

Arbitrary data may be associated with definitions. This mechanism is used by `arglist*`, `docstring*` and `source-location*` for easy extension.

The following functions take an `xref` argument and not a `dref(0 1)` to allow working with `non-canonical` or non-existent definitions.

- **[function]** `definition-property` *obj indicator*

Return the value of the property associated with *obj* whose name is `eql` to *indicator*. The second return value indicates whether the property was found. `setfable`.

- *Regular definition*: *obj* is commonly an `xref(0 1)`. `xrefs` that are `xref=` are equivalent for the purposes of `definition-property`.
- *Irregular case*: If *obj* is not an `xref` (and, by extension, not a `dref(0 1)`), it is equivalent to other objects to which it is `equal`.

For example, see the `docstring*` generic-function, which uses both the regular and the irregular cases.

- **[function]** `delete-definition-property` *obj indicator*

Delete the property *indicator* of *obj* established by setting `definition-property`. Return true if the property was found.

- **[function]** `definition-properties` *xref*

Return the properties of *xref* as an association list.

- **[function]** `delete-definition-properties` *xref*

Delete all properties associated with *xref*.

- **[function]** `move-definition-properties` *from-xref to-xref*

Associate all properties of `from-xref` with `to-xref`, as if readding them one-by-one with `(setf definition-property)`, and deleting them from `from-xref` with `delete-definition-property`.

## 9.6 dref-classes

These are the `dref-classes` corresponding to [Basic Locative Types](#). They are exported to make it possible to go beyond the [Basic Operations](#) (e.g. `pax:document-object*`). For [Defining Locative Types](#), they are not necessary, as `define-locative-type` handles inheritance automatically based on its `locative-supertypes` argument.

### for Variables

- `[class] variable-dref` *dref*  
`dref-ext:dref-class` of `variable`.
- `[class] constant-dref` *variable-dref*  
`dref-ext:dref-class` of `mgl-pax:constant`.

### for Macros

- `[class] macro-dref` *dref*  
`dref-ext:dref-class` of `mgl-pax:macro`.
- `[class] symbol-macro-dref` *dref*  
`dref-ext:dref-class` of `mgl-pax:symbol-macro`.
- `[class] compiler-macro-dref` *dref*  
`dref-ext:dref-class` of `compiler-macro`.
- `[class] setf-dref` *dref*  
`dref-ext:dref-class` of `setf`.
- `[class] setf-compiler-macro-dref` *compiler-macro-dref*  
`dref-ext:dref-class` of `dref:setf-compiler-macro`.

### for Functions

- `[class] function-dref` *dref*  
`dref-ext:dref-class` of `function`.
- `[class] setf-function-dref` *function-dref setf-dref*  
`dref-ext:dref-class` of `dref:setf-function`.
- `[class] generic-function-dref` *function-dref*  
`dref-ext:dref-class` of `generic-function`.

- [class] `setf-generic-function-dref` *generic-function-dref setf-function-dref*  
dref-ext:dref-class of dref:setf-generic-function.
- [class] `method-dref` *dref*  
dref-ext:dref-class of method.
- [class] `setf-method-dref` *method-dref setf-dref*  
dref-ext:dref-class of dref:setf-method.
- [class] `method-combination-dref` *dref*  
dref-ext:dref-class of method-combination.
- [class] `reader-dref` *method-dref*  
dref-ext:dref-class of mgl-pax:reader.
- [class] `writer-dref` *method-dref*  
dref-ext:dref-class of mgl-pax:writer.
- [class] `accessor-dref` *reader-dref writer-dref setf-method-dref*  
dref-ext:dref-class of mgl-pax:accessor.
- [class] `structure-accessor-dref` *setf-function-dref function-dref*  
dref-ext:dref-class of mgl-pax:structure-accessor.

#### for Types and Declarations

- [class] `type-dref` *dref*  
dref-ext:dref-class of type.
- [class] `class-dref` *type-dref*  
dref-ext:dref-class of class.
- [class] `declaration-dref` *dref*  
dref-ext:dref-class of declaration.

#### for the Condition System

- [class] `condition-dref` *class-dref*  
dref-ext:dref-class of condition.
- [class] `restart-dref` *symbol-locative-dref*  
dref-ext:dref-class of restart.

#### for Packages and Readtables

- [class] `asdf-system-dref` *dref*  
dref-ext:dref-class of asdf/system:system.

- **[class]** `package-dref` *dref*  
[dref-ext:dref-class](#) of `package`.
- **[class]** `readtable-dref` *dref*  
[dref-ext:dref-class](#) of `readtable`.

#### for Unknown Definitions

- **[class]** `unknown-dref` *dref*  
[dref-ext:dref-class](#) of `mgl-pax:unknown`.

#### for DRef Constructs

- **[class]** `dtype-dref` *dref*  
[dref-ext:dref-class](#) of `dref:dtype`.
- **[class]** `locative-dref` *dtype-dref*  
[dref-ext:dref-class](#) of `mgl-pax:locative`.
- **[class]** `symbol-locative-dref` *dref*  
 All `locative` types defined with `define-symbol-locative-type` inherit from this class.
- **[class]** `lambda-dref` *dref*  
[dref-ext:dref-class](#) of `lambda`.

## 9.7 Source Locations

These represent the file or buffer position of a **defining form** and are returned by the [source-location](#) function. For the details, see the `Elisp` function `slime-goto-source-location`.

- **[function]** `make-source-location` *&key file file-position buffer buffer-position snippet*  
 Make a Swank source location. The ultimate reference is `slime-goto-source-location` in `slime.el`. When `snippet` is provided, the match nearest to `file-position` is determined (see the `Elisp` `slime-isearch` and [source-location-adjusted-file-position](#)).
- **[function]** `source-location-p` *object*  
 See if `object` is a source location object.
- **[function]** `source-location-file` *location*  
 Return the name of the file of the **defining form**. This may be `nil`, for example, if `location` is of a **defining form** that was entered at the REPL, or compiled in the `*slime-scratch*` buffer.
- **[function]** `source-location-file-position` *location*  
 Return the file position of the **defining form** or `nil` if it's not available. The first position is 0.

- **[function]** `source-location-buffer` *location*

Return the name of the Emacs buffer of the **defining form** or `nil` if there is no such Emacs buffer.

- **[function]** `source-location-buffer-position` *location*

Return the position of the **defining form** in `source-location-buffer` or `nil` if it's not available. The first position is 1.

- **[function]** `source-location-snippet` *location*

Return the **defining form** or a prefix of it as a string or `nil` if it's not available.

- **[function]** `source-location-adjusted-file-position` *location*

Return the actual file position *location* points to allowing for some deviation from the raw `source-location-file-position`, which is adjusted by searching for the nearest occurrence of `source-location-snippet` in the file. The file is read using the same external format that ASDF would use to compile it. Needless to say, this can be a very expensive operation.

If `source-location-file` is `nil`, `nil` is returned. If there is no snippet, or it doesn't match, then `source-location-file-position` (or 0 if that's `nil`) is returned.

This is a non-interactive companion to the Elisp function `slime-location-offset`, supporting only file positions and non-partial matching of snippets.

- **[macro]** `this-source-location`

The value of this macro form is a function of no arguments that returns its own `source-location`.

## 10 Indices

Referrer definition type abbreviations:

- *f*: for definitions in the function namespace (macros, compiler macros and also methods)
- *t*: DEFTYPEs, classes, conditions, structs
- *d*: documentation sections and glossary terms
- *l*: definitions of definition types
- *s*: ASDF systems
- *p*: packages
- *n*: named readtables
- *v*: special variables and constants
- *r*: restarts
- *?*: other

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